MIL-M-28787/136A(EC)
12 June 1979
SUPERSEDING
MIL-M-28787/136(NAVY)
29 July 1974

MILITARY SPECIFICATION

* MODULES, STANDARD ELECTRONIC

TRANSFORMER, POWER, KEY CODE RPN

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic power transformer module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Transformer
Family	Power
Key code	RPN
Size	1U
Weight	2.27 kg maximum
Failure rate	0.865 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Working voltage	225 V peak
Input voltage	110 to 121 Vrms, 400 Hz ±6%
Maximum power dissipation	24 W
Commutating inductance	
Secondary No. 1 and No. 2	36 µH
Secondary No. 3 and No. 4	4.7 µH
Core loss	16 W maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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* 1.3 Module function. This module contains a transformer with one primary and six secondary windings. The primary is a three-phase, three-wire open-delta connection; three of the secondaries are wye-connected each with a neutral, the fourth secondary is wye-connected without a neutral, and the remaining two are single-phase windings. Each primary winding has two taps and all secondary windings, except those of secondary No. 1, have one tap. There is an electrostatic shield between the primary and secondaries, the shield is tied to the core.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-27 - Transformers and Inductors (Audio, Power, and High-Power Pulse), General Specification for.
MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtainedfrom the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.865 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be RPN.

3.5.2 Size. The module size shall be 1U.

3.5.3 Weight. The module shall have a maximum weight of 2.27 kg.

3.5.4 Contact pins. The module shall have 80 contact pins positioned A1 through B40.

3.5.5 Fin position. The module shall have an insertion and extraction on increments A and T, and heat dissipation fin on increments B through U.

3.5.6 Depth. The module shall have additional depth requirements per figure 5 of MIL-STD-1389 for increments C through U.

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3.5.7 Holddown holes. The module shall have additional requirements for location (increments C and D) and quantity of holddown holes (two holes, two places).

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/136-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Transformer requirements. Each module transformer shall conform to the requirements of MIL-T-27, type TF5V03ZZ, except as modified herein.

3.6.1 Seal test. The immersion sealing test of MIL-T-27 shall not be performed.

3.7 Overvoltage operation. The transformer shall be capable of 30 minutes operation with 150 Vrms line-to-line applied to the 115 Vrms primary tap connections without damage to the unit.

3.8 Circuit pin configuration. The function diagram and pin assignments shall conform to figures 1 and 2.

3.9 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.10 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below.

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
Digital multifunction meter	Item M02	2
Megohm bridge	Item R06	1
Power sine wave generator	Item G09	1
Phase angle voltmeter	Item Z01	1
Isolation transformer	Item E01	1
Ratio transformer	Item F02	1
Power amplifier	Item A01	1
AC wattmeter 0-150 V, 0-2 A, 400 Hz		2
Resistors		
0.1Ω ±1%, 5 W		8
Load bank (LB) 0.8 to 16Ω (4.25 A)		6
Load bank (LB) 0.8 to 10Ω (4.25 A)		6
Load bank (LB) 100 to 500Ω (0.1A)		2
100Ω ±1%, 5 W		2
0.1Ω ±1%, 10 W		1

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

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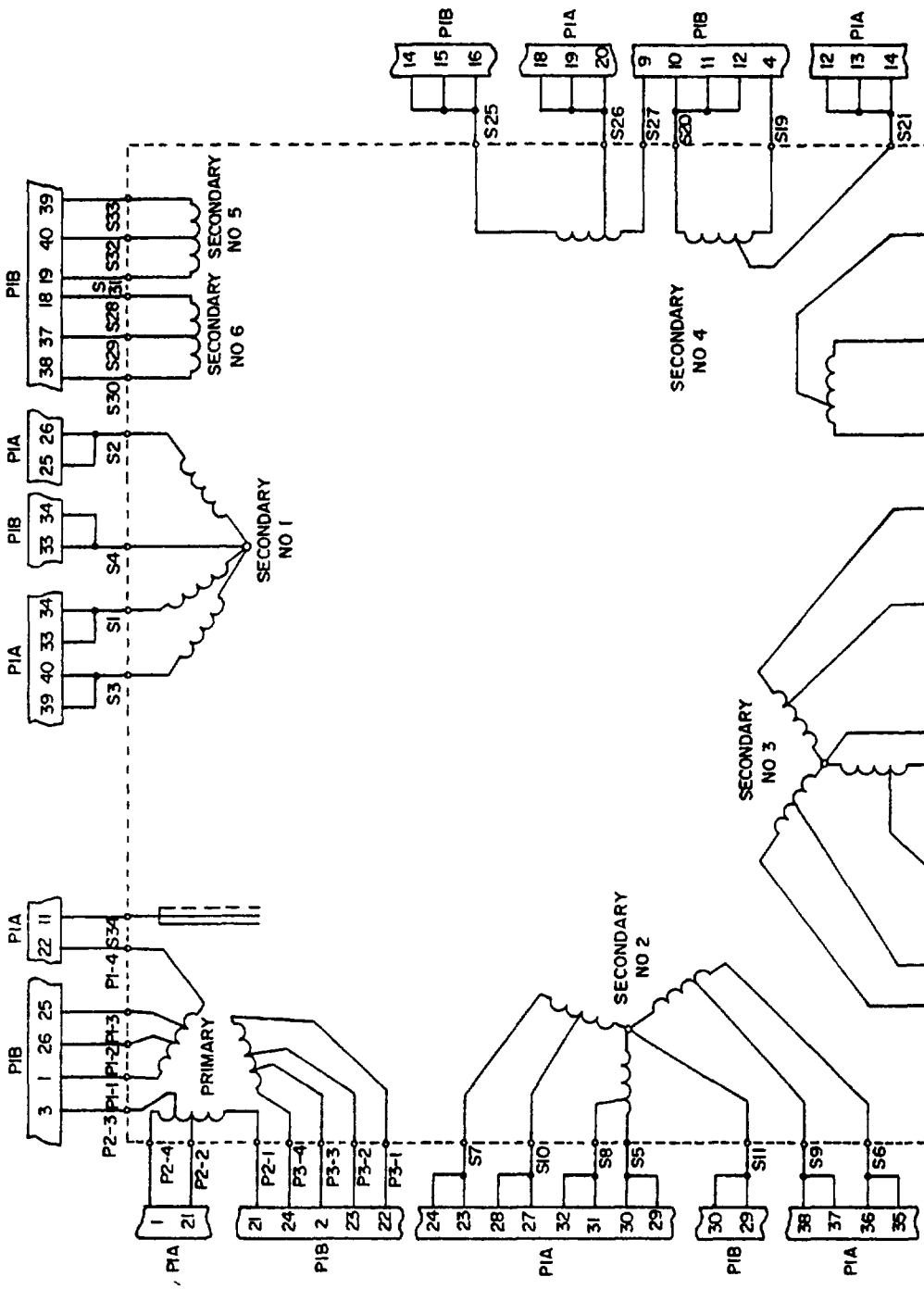
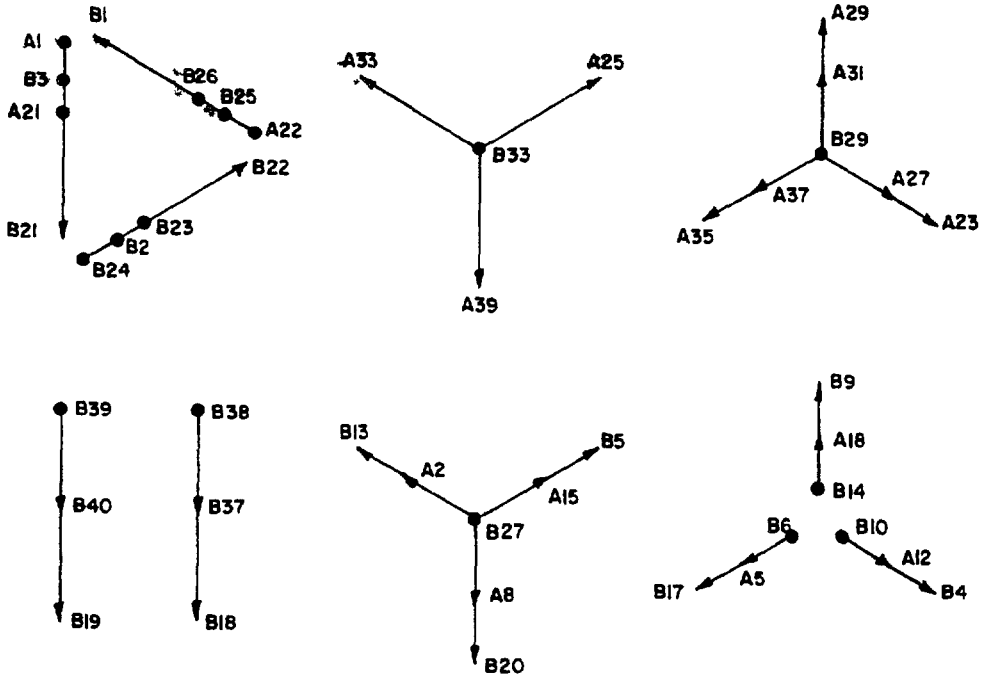


FIGURE 1. RPN module function diagram, transformer.

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NOTE For additional pins see figure 1 (module schematic).

FIGURE 2. RPN module phasor diagram.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/, 2/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A32	Jumper continuity	II	X	0	0.1	0	0.1	0	0.1	0	0.1	Ω
A33	Continuity	II	S	0	1.0	0	1.0	0	1.0	0	1.0	Ω
B1-B11	Insulation resistance	III	X	10	∞	10	∞	10	∞	10	∞	MΩ
B12	Module integrity	III	X	10	∞	10	∞	10	∞	10	∞	MΩ
C1-C6	Transformation ratio (no load)	IV		0.16874	0.17214	0.16874	0.17214	0.16874	0.17214	0.16874	0.17214	Ratio
C7-C9				0.16098	0.16424	0.16098	0.16424	0.16098	0.16424	0.16098	0.16424	
C10-C12				0.06543	0.06675	0.06543	0.06675	0.06543	0.06675	0.06543	0.06675	
C13 C15				0.04391	0.04479	0.04391	0.04479	0.04391	0.04479	0.04391	0.04479	
C16-C18				0.05854	0.05972	0.05854	0.05972	0.05854	0.05972	0.05854	0.05972	
C19-C21				0.04391	0.04479	0.04391	0.04479	0.04391	0.04479	0.04391	0.04479	
C22-C25	Transformation ratio (no load)	IV	X	0.18251	0.18619	0.18251	0.18619	0.18251	0.18619	0.18251	0.18619	Ratio

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/, 2/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
* C26-C28	Transformation ratio (no load)	IV	X	0.17658	0.18014	0.17658	0.18014	0.17658	0.18014	0.17658	0.18014	Ratio
C29-C31	Transformation ratio (no load)	IV		0.16041	0.16365	0.16041	0.16365	0.16041	0.16365	0.16041	0.16365	
D1-D6	Transformation ratio (loaded)	V		0.16522	-	0.16522	-	0.16418	-	0.16418	-	
* D7-D9				0.10087	-	0.10087	-	0.10032	-	0.09987	-	
D10-D12				0.15764	-	0.15764	-	0.15665	-	0.15665	-	
* D13-D15				0.10870	-	0.10870	-	0.10827	-	0.10800	-	
* D16-D21				0.04304	-	0.04304	-	0.04278	-	0.04262	-	
* D22-D25				0.17913	-	0.17913	-	0.17807	-	0.17781	-	
D26-D28	Transformation ratio (loaded)	V	X	0.15703	-	0.15703	-	0.15604	-	0.15604	-	Ratio

See footnotes at end of table.

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TABLE I. Module electrical requirements. - Continued

Test condition	Parameter	Ref test table	Test type 1/, 2/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
* D29-D31	Transformation ratio (loaded)	V	X	0.17273	-	0.17273	-	0.17162	-	0.17153	-	Ratio
F1-F6	Commutating inductance	VI	S	0	36	0	36	0	36	0	36	µH
F7-F12	Commutating inductance	VI	S	0	4.7	0	4.7	0	4.7	0	4.7	µH
G1a-G1c	Core loss	VII	S	0	16	0	16	0	16	0	16	W
H1-H3	Induced voltage (current)	VIII	S	0	25	0	25	0	25	0	25	%Δ
J1A1-J1A32	Jumper resistance	IX	S	0	0.020	0	0.025	0	0.025	0	0.030	Ω

1/ All "X" parameters shall be 100 percent tested during module acceptance tests.
2/ All "S" parameters shall be 25°C ambient sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater).

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4.4.2 Order of testing. Test conditions A through J shall be performed in any order with the following exceptions. If test condition B is part of a sequence of tests, the order of testing shall be as follows: B, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through J in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through C, G, and H, shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. The module shall be nonoperating during the test. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The life test shall be run with alternate cycles of the test circuits shown on figures 3 and 4. The test shall start with the circuit of figure 3 (that is, figure 3 first cycle; figure 4 second cycle; figure 3 third cycle, and so forth). Following the return of the module to 25°C, electrical tests B, C, D, G, H, and J shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

4.6 Overvoltage operation. The test setup shall be in accordance with figure 5. Each lead shall be adjusted for 4.25 A ±3%. Operate the transformer for 30 minutes. At the end of this period, the transformer shall conform to the insulation resistance and induced voltage requirements specified in table I (test conditions B and H).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.10).

6.3 Cross reference of classification. Part number M28787/136-1 supersedes part number M28787/136. The two parts are interchangeable.

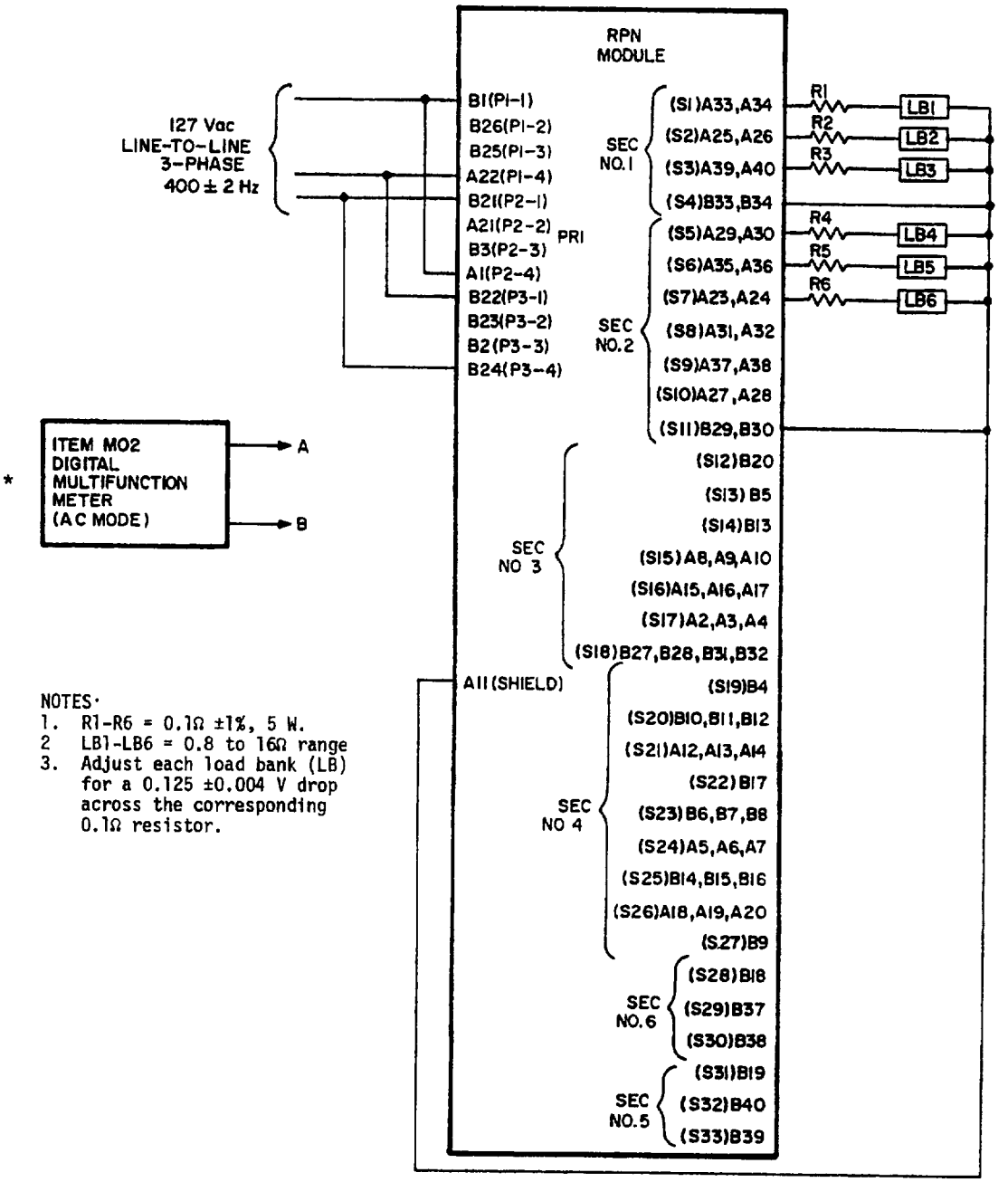
* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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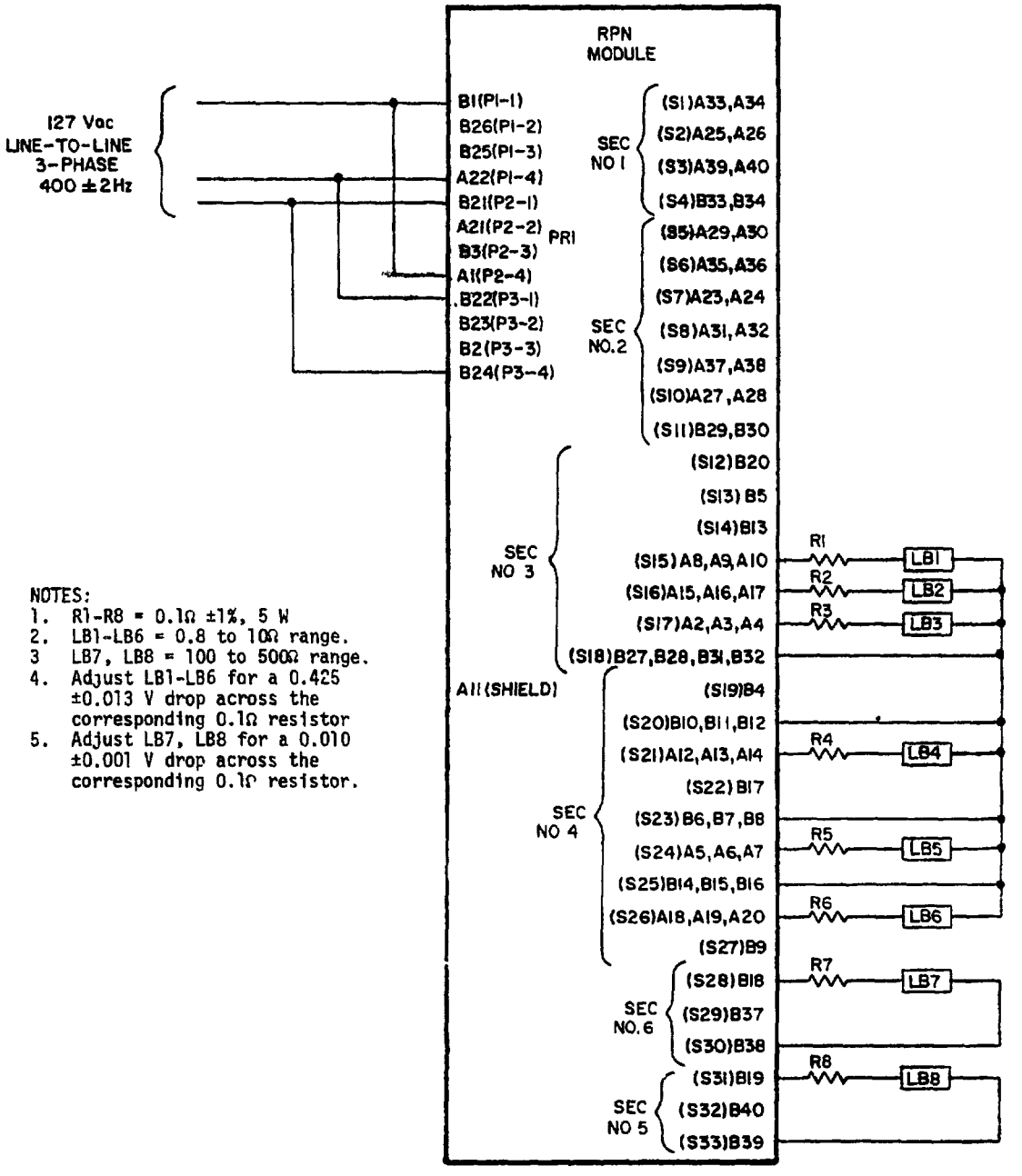
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* NOTES:
1. R1-R6 = 0.1Ω ±1%, 5 W.
2 LB1-LB6 = 0.8 to 16Ω range
3. Adjust each load bank (LB) for a 0.125 ±0.004 V drop across the corresponding 0.1Ω resistor.

FIGURE 3. Operating temperature and life test (1) circuit.

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NOTES:
1. R1-R8 = 0.1Ω ±1%, 5 W
2. LB1-LB6 = 0.8 to 10Ω range.
3 LB7, LB8 = 100 to 500Ω range.
4. Adjust LB1-LB6 for a 0.425 ±0.013 V drop across the corresponding 0.1Ω resistor
5. Adjust LB7, LB8 for a 0.010 ±0.001 V drop across the corresponding 0.1Ω resistor.

FIGURE 4. Life test (2) circuit.

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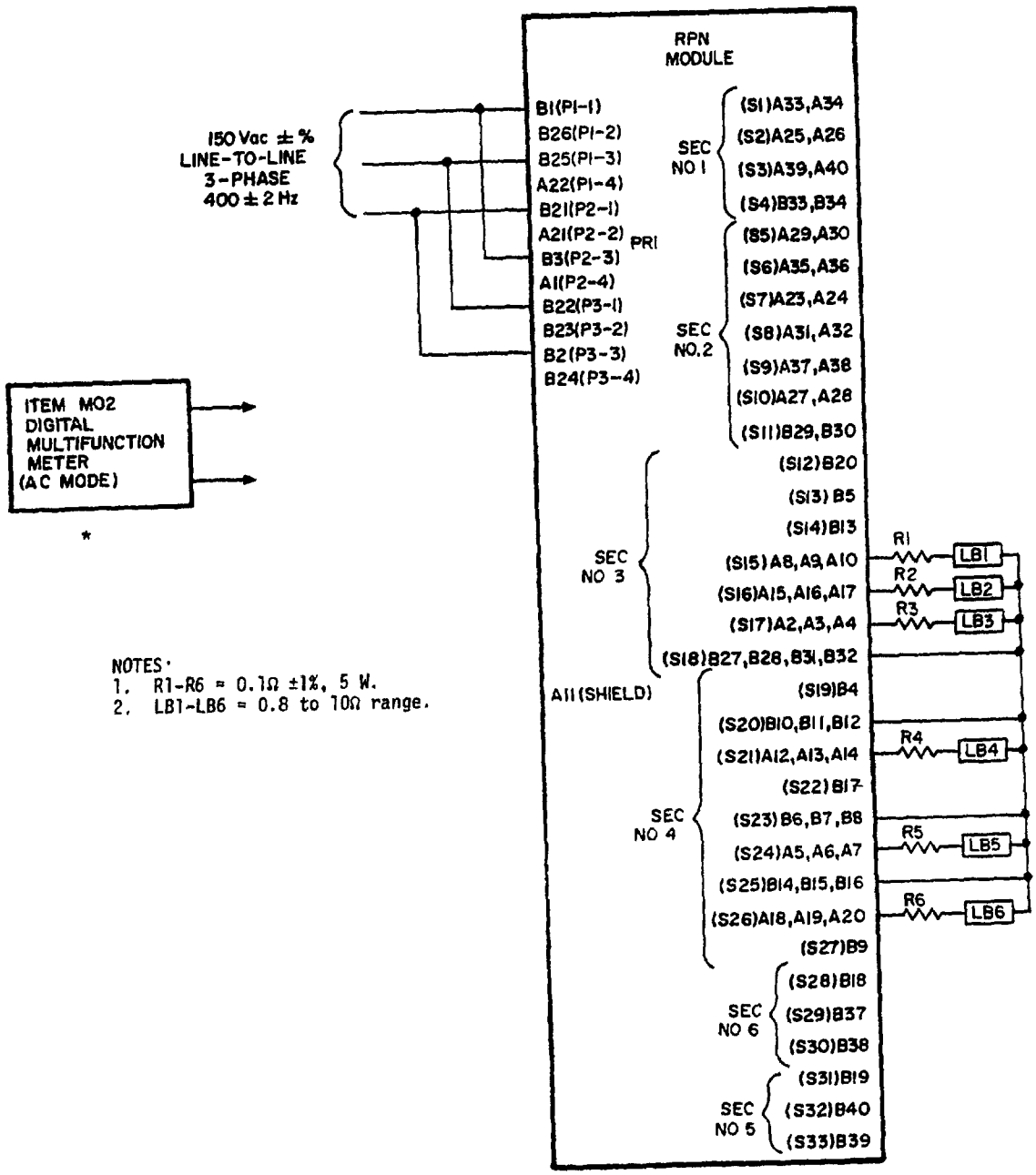


FIGURE 5. Overvoltage test circuit.

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TABLE II. Jumper continuity test.

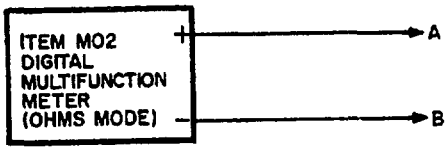
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
A1	Jumper continuity, secondary	2-S7	A23	A24	0	0.1	Ω	6
A2		1-S2	A25	A26				
A3		2-S10	A27	A28				
A4		2-S11	B29	B30				
A5		2-S8	A31	A32				
A6		1-S4	B33	B34				
A7		2-S6	A35	A36				
A8		2-S9	A37	A38				
A9		1-S3	A39	A40				
A10		3-S17	A2	A3				
A11		3-S17	A2	A4				
A12		4-S24	A5	A6				
A13		4-S24	A5	A7				
A14		3-S15	A8	A9				
A15		3-S15	A8	A10				
A16		4-S21	A12	A13				
A17		4-S21	A12	A14				
A18	Jumper continuity, secondary	3-S16	A15	A16	0	0.1	Ω	6

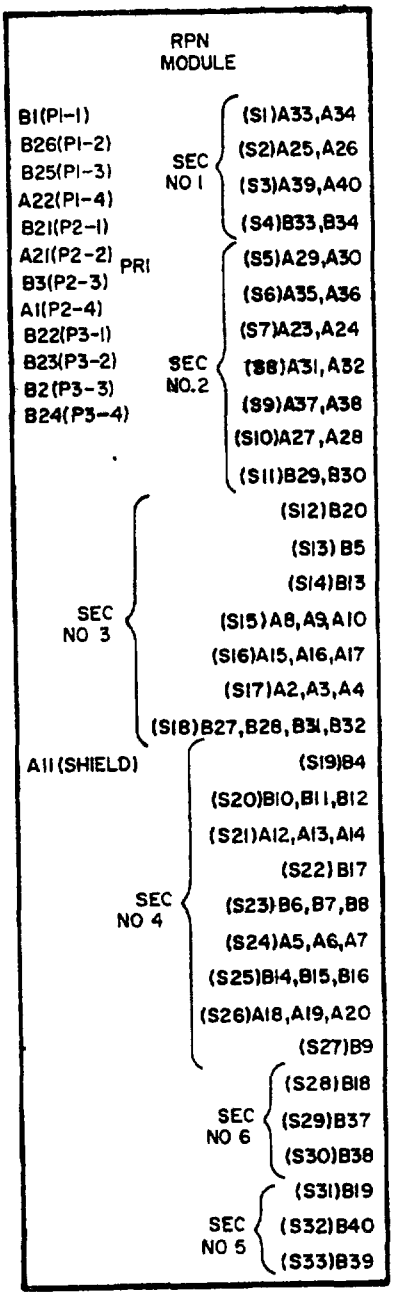
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TABLE II. Jumper continuity test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
A19	Jumper continuity, secondary	3-S16	A15	A17	0	0.1	Ω	6
A20		4-S26	A18	A19				
A21		4-S26	A18	A20				
A22		1-S4	B33	B34				
A23		4-S23	B6	B7				
A24		4-S23	B6	B8				
A25		4-S20	B10	B11				
A26		4-S20	B10	B12				
A27		3-S18	B27	B28				
A28		3-S18	B27	B31				
A29		3-S18	B27	B32				
A30		2-S11	B29	B30				
A31		4-S25	B14	B15				
A32	Jumper continuity, secondary	4-S25	B14	B16		0.1		
A33	Continuity		A11	A11 fins	0	1.0	Ω	6

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* NOTES:
1. Readings shall be compensated for test circuit resistance that are in excess of 0.010Ω
2. For test condition A33, penetrate the anodized finish on the module fins with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Jumper continuity and resistance test circuit for tables II and IX.

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TABLE III. Insulation resistance and module integrity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Insulation resistance P1-4 and P1-1	A22 B1	A1,A11,A23,A25, A29,A33,A35,A39, B4,B5,B7,B9,B11, B13,B15,B17-B22, B24,B38,B39	10	∞	MΩ	7
B2	P2-4 and P2-1	A1 B21	A11,A22,A23,A25, A29,A33,A35,A39, B1,B4,B5,B7,B9, B11,B13,B15,B17- B20,B22,B24,B38, B39				
B3	P3-1 and P3-4	B22 B24	A1,A11,A22,A23, A25,A29,A33,A35, A39,B1,B4,B5,B7, B9,B11,B13,B15, B17,B18,B19,B20, B21,B38,B39				
B4	Secondary No. 5	B19 B39	A1,A11,A22,A23, A25,A29,A33,A35, A39,B1,B4,B5,B7, B9,B11,B13,B15, B17,B18,B20,B21, B22,B24,B38				
B5	Insulation resistance Secondary No. 6	B18 B38	A1,A11,A22,A23, A25,A29,A33,A35, A39,B1,B4,B5,B7, B9,B11,B13,B15, B17,B19,B20,B21, B22,B24,B39	10	∞	MΩ	7

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TABLE III. Insulation resistance and module integrity test - Continued.

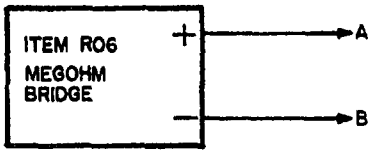
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B6	Insulation resistance Secondary No. 1	A25 A33 A39	A1,A11,A22,A23, A29,A35,B1,B4, B5,B7,B9,B11, B13,B15,B17,B18, B19,B20,B21,B22, B24,B38,B39	10	-	MΩ	7
B7	Secondary No. 2	A23 A29 A35	A1,A11,A22,A25, A33,A39,B1,B4, B5,B7,B9,B11, B13,B15,B17,B18, B19,B20,B21,B22, B24,B38,B39				
B8	Secondary No. 3	B5 B13 B20	A1,A11,A22,A23, A25,A29,A33,A35, A39,B1,B4,B7,B9, B11,B15,B17,B18, B19,B21,B22,B24, B38,B39				
B9	Secondary No. 4	B7 B17	A1,A11,A22,A23, A25,A29,A33,A35, A39,B1,B4,B5,B9, B11,B13,B15,B18, B19,B20,B21,B22, B24,B38,B39				
B10	Insulation resistance Secondary No. 4	B9 B15	A1,A11,A22,A23, A25,A29,A33,A35, A39,B1,B4,B5,B7, B11,B13,B17,B18, B19,B20,B21,B22, B24,B38,B39	10	-	MΩ	7

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TABLE III. Insulation resistance and module integrity test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B11	Insulation resistance Secondary No. 4	B4 B11	A1,A11,A22,A23, A25,A29,A33,A35, A39,B1,B5,B7,B9, B13,B15,B17,B18, B19,B20,B21,B22, B24,B38,B39	10	∞	MΩ	7
B12	Module integrity	B35 B36	A1,A11,A22,A23, A25,A29,A33,A35, A39,B1,B4,B5,B7, B9,B11,B13,B15, B17,B18,B19,B20, B21,B22,B24,B38, B39	10	∞	MΩ	7

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NOTES:
1. Megohm bridge voltage for B1-B3 = 500 Vdc ±5%.
2. Megohm bridge voltage for B4-B12 = 100 Vdc ±5%.

WARNING:

When using the megohm bridge, high voltage may be present at any one of the output binding posts. Always set the function switch of the megohm bridge to discharge before connecting or disconnecting the test leads.

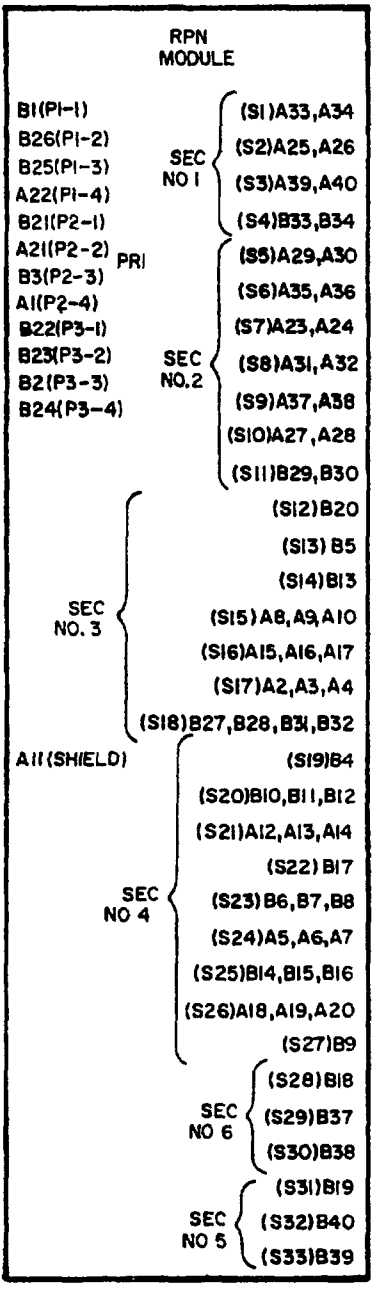


FIGURE 7. Insulation resistance and module integrity test circuit for table III.

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TABLE IV. Transformation ratio (no load) test.

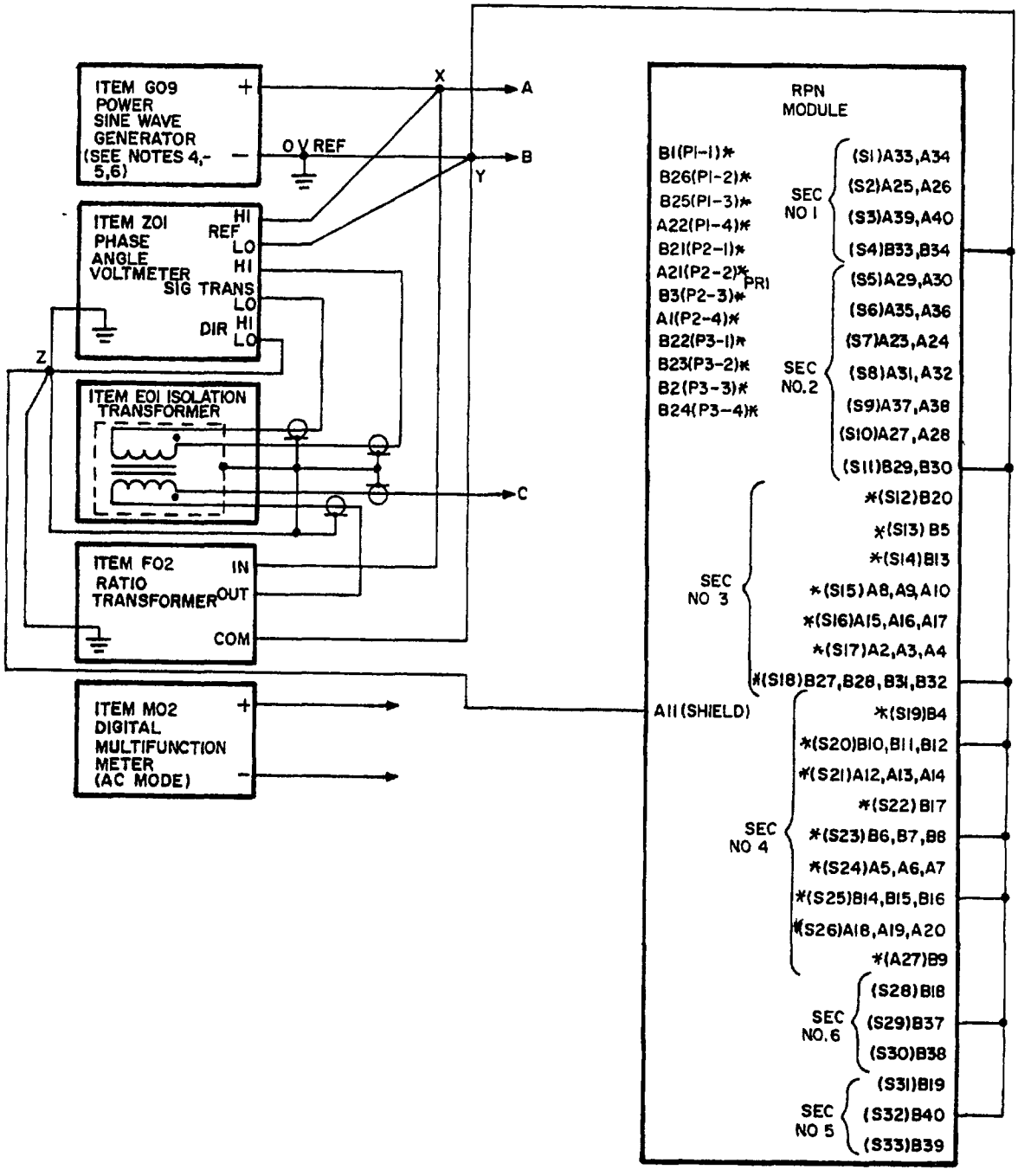
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	Transformation ratio (no load)	B21	B3	A40	0.16874	0.17214	Ratio	8
C2		B1	B25	A34				
C3		B22	B2	A26				
C4		B3	B21	A30				
C5		B25	B1	A24				
C6		B2	B22	A36	0.16874	0.17214		
C7		B3	B21	A32	0.16098	0.16424		
C8		B25	B1	A28	0.16098	0.16424		
C9		B2	B22	A38	0.16098	0.16424		
C10		B21	B3	B20	0.06543	0.06675		
C11		B1	B25	B13	0.06543	0.06675		
C12		B22	B2	B5	0.06543	0.06675		
C13		B21	B3	A8	0.04391	0.04479		
C14		B1	B25	A2	0.04391	0.04479		
C15		B22	B2	A15	0.04391	0.04479		
C16		B25	B1	B4	0.05854	0.05972		
C17		B2	B22	B17	0.05854	0.05972		
C18		B3	B21	B9	0.05854	0.05972		
C19		B25	B1	A12	0.04391	0.04479		
C20		B3	B21	A18	0.04391	0.04479		
C21	Transformation ratio (no load)	B2	B22	A5	0.04391	0.04479	Ratio	8

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TABLE IV. Transformation ratio (no load) test. - Continued

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C22	Transformation ratio (no load)	B21	B3	B19	0.18251	0.18619	Ratio	8
C23		B3	B21	B39	0.18251	0.18619		
C24		B21	B3	B18	0.18251	0.18619		
C25		B3	B21	B38	0.18251	0.18619		
C26		B21	A21	A40	0.17658	0.18014		
C27		B22	B23	A26	0.17658	0.18014		
C28		B1	B26	A34	0.17658	0.18014		
C29		B1	A22	A34	0.16041	0.16365		
C30		B21	A1	A40	0.16041	0.16365		
C31	Transformation ratio (no load)	B22	B24	A26	0.16041	0.16365	Ratio	8

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* FIGURE 8. Transformation ratio (no load) test circuit for table IV.

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NOTES:
1. All connections made at points X, Y and Z shall terminate at one point as shown.
2. All pins jumpered together internal to the module (i.e. pins A33, A34) shall also be jumpered external to the module directly at the transformer mating connector.
3. *Denotes No. 24 or larger external jumper wire between common pins and interconnecting circuits wire; all other wires No. 30 or larger.
4. For test conditions C1-C25, E_{IN} = 115 Vrms ±1% at 400 ±2 Hz.
5. For test conditions C26-C28, E_{IN} = 110 Vrms ±1% at 400 ±2 Hz.
6. For test conditions C29-C31, E_{IN} = 121 Vrms ±1% at 400 ±2 Hz.

FIGURE 8. Transformation ratio (no load) test circuit for table IV - Continued.

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TABLE V. Transformation ratio (loaded) test.

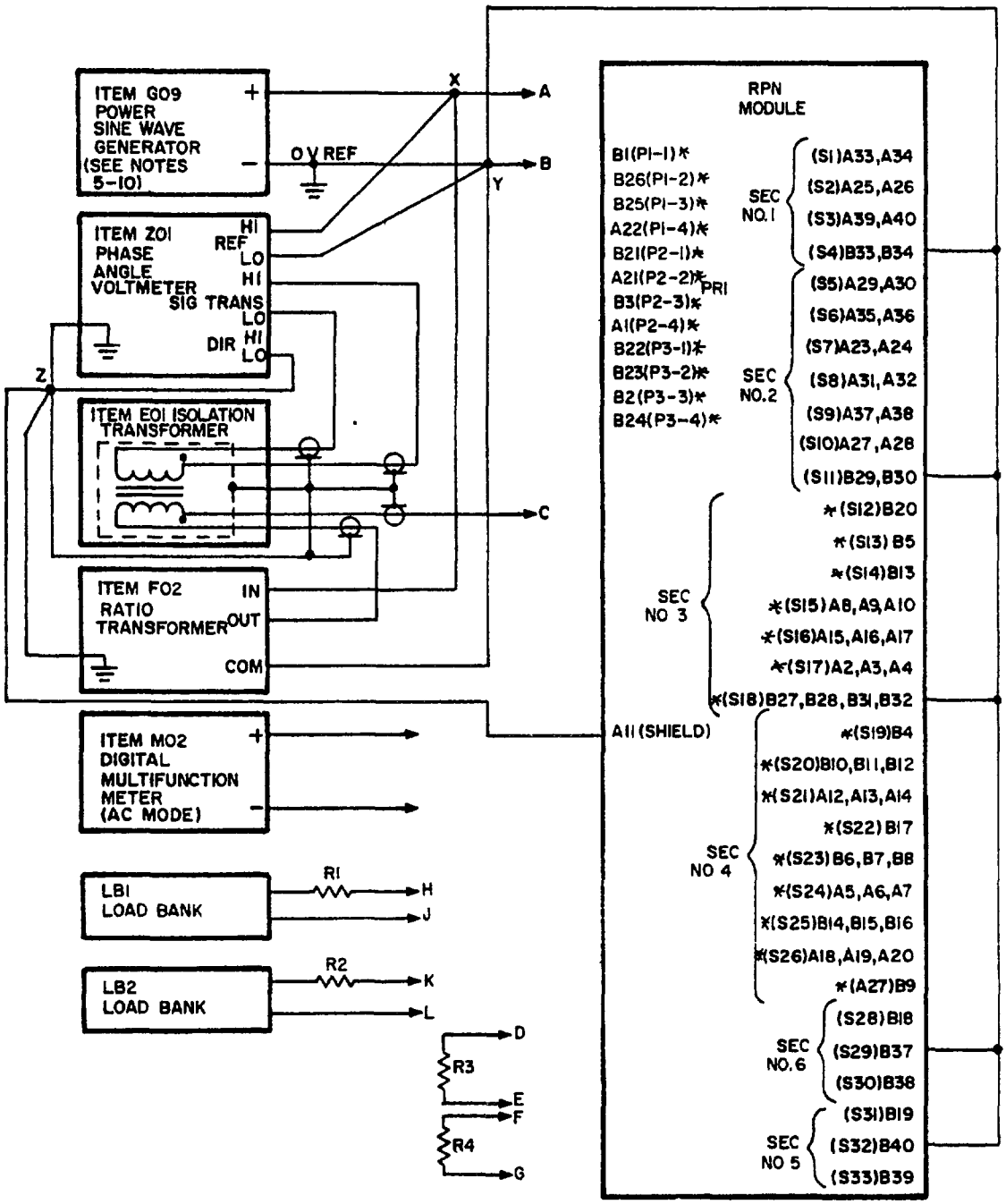
Test condition	Parameter	Connect											Initial limits Ambient 25° ±5°C		Unit	Test figure	Notes
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	J to pin	K to pin	L to pin	Min	Max			
D1	Transformation ratio (loaded)	B1	B25	A34	-	-	-	-	A33	B33	B11	B27	0.16522	-	Ratio	9	5
D2		B22	B2	A26	-	-	-	-	A25	B33	B7						
D3		B21	B3	A40	-	-	-	-	A39	B33	B15						
D4		B3	B21	A30	-	-	-	-	A29	B29	B15						
D5		B2	B22	A36	-	-	-	-	A35	B29	B7						
D6		B25	B1	A24	-	-	-	-	A23	B29	B11		0.16522				
D7		B1	B25	B10	-	-	-	-	A33	B33	B11		0.10087				
D8		B22	B2	B6	-	-	-	-	A25	B33	B7		0.10087				
D9		B21	B3	B14	-	-	-	-	A39	B33	B15		0.10087				5
D10		B25	B1	A28	-	-	-	-	A27	B29	B11		0.15764				6
D11		B2	B22	A38	-	-	-	-	A37	B29	B7		0.15764				6
D12		B3	B21	A32	-	-	-	-	A31	B29	B15		0.15764				6
D13		B1	B25	B10	-	-	-	-	A33	B33	B11		0.10870				7
D14		B22	B2	B6	-	-	-	-	A25	B33	B7		0.10870				7
D15		B21	B3	B14	-	-	-	-	A39	B33	B15	B27	0.10870				7
D16		B1	B25	A2	-	-	-	-	A3	B31	A13	B11	0.04304				8
D17		B25	B1	A12	-	-	-	-	A3		A13	B11					
D18		B22	B2	A15	-	-	-	-	A16		A6	B7					
D19		B2	B22	A5	-	-	-	-	A16		A6	B7					
D20	Transformation ratio (loaded)	B21	B3	A8	B19	B40	B18	B37	A9	B31	A19	B15	0.04304	-	Ratio	9	8

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TABLE V. Transformation ratio (loaded) test - Continued.

Test condition	Parameter	Connect											Initial limits Ambient 25° ±5°C		Unit	Test figure	Notes
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	J to pin	K to pin	L to pin	Min	Max			
D21	Transformation ratio (loaded)	B3	B21	A18	B39	B40	B38	B37	A9	B31	A19	B15	0.04304	-	Ratio	9	8
D22		B3	B21	B38	B39		B38						0.17913				
D23		B3	B21	B39	B39		B38						0.17913				
D24		B21	B3	B18	B19		B18						0.17913				
D25		B21	B3	B19	B19	B40	B18	B37	A9	B31	A19	B15	0.17913				8
D26		A22	B1	A24	-	-	-	-	A23	B29	B11	B27	0.15703				9
D27		B24	B22	A36	-	-	-	-	A35	B29	B7	B27	0.15703				9
D28		A1	B21	A30	-	-	-	-	A29		B15		0.15703				9
D29		A21	B21	A30	-	-	-	-	A29		B15		0.17273				10
D30		B23	B22	A36	-	-	-	-	A35		B7		0.17273				10
D31	Transformation ratio (loaded)	B26	B1	A24	-	-	-	-	A23	B29	B11	B27	0.17273	-	Ratio	9	10

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* FIGURE 9. Transformation ratio (loaded) test circuit for table V

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NOTES:
1. R1, R2 = 0.1Ω ±1%, 5 W.
2. R3, R4 = 100Ω ±1%, 5 W.
3. All pins jumpered together internal to the module (i.e. pins A33, A34) shall also be jumpered external to the module directly at the transformer mating connector.
4. *Denotes No. 24 or larger jumper wire between common pins and interconnecting circuits; all other wires, No. 30 or larger.
5 Additional (test setup) parameters for test condition D1 through D9:

   a. Jumper pins A3 to B4, A16 to B17 and A9 to B9.
   b. E_{IN} = 115 Vrms ±1%, 400 ±2 Hz.
   c. Load 1 current = 2.07 amperes ±3%.
   d. Load 2 current = 2.07 amperes ±3%.

6. Additional (test setup) parameters for test conditions D10, D11 and D12:

   a. Jumper pins A3 to B4, A16 to B17 and A9 to B9.
   b. E_{IN} = 115 Vrms ±1%, 400 ±2 Hz.
   c. Load 1 current = 1.98 amperes ±3%.
   d. Load 2 current = 2.07 amperes ±3%

7. Additional (test setup) parameters for test conditions D13, D14 and D15·

   a. Jumper pins A13 to B13, A6 to B5 and A19 to B20.
   b. E_{IN} = 115 Vrms ±1%, 400 ±2 Hz.
   c Load 1 current = 2.07 amperes ±3%.
   d. Load 2 current = 2.07 amperes ±3%.

8 Additional (test setup) parameters for test conditions D16 through D25:

   a. Jumper pins B6, B10, B14 and B28.
   b. E_{IN} = 115 Vrms ±1%, 400 ±2 Hz.
   c. Load 1 current = 4.125 amperes ±3%.
   d. Load 2 current = 4.125 amperes ±3%.
9. Additional (test setup) parameters for test conditions D26, D27 and D28:

   a Jumper pins A3 to B4, A16 to B17 and A9 to B9
   b. E_{IN} = 121 Vrms ±1%, 400 ±2 Hz.
   c. Load 1 current = 2.07 amperes ±3%
   d. Load 2 current = 2.07 amperes ±3%.

10. Additional test (setup) parameters for test conditions D29, D30 and D31.

   a. Jumper pins A3 to B4, A16 to B17 and A9 to B9.
   b. E_{IN} = 110 Vrms ±1%, 400 ±2 Hz.
   c. Load 1 current = 2.07 amperes ±3%.
   d. Load 2 current = 2.07 amperes ±3%.

11. All connections made at points X, Y and Z shall terminate at one point as shown.
12. LB1-LB2 = 0.8 to 10Ω range.
13. Adjust each load bank to the correct load current as specified in notes 5 through 10 by monitoring the voltage across the corresponding 0.1Ω resistor. Adjustment shall be made until voltage = (0.1) (required load current).

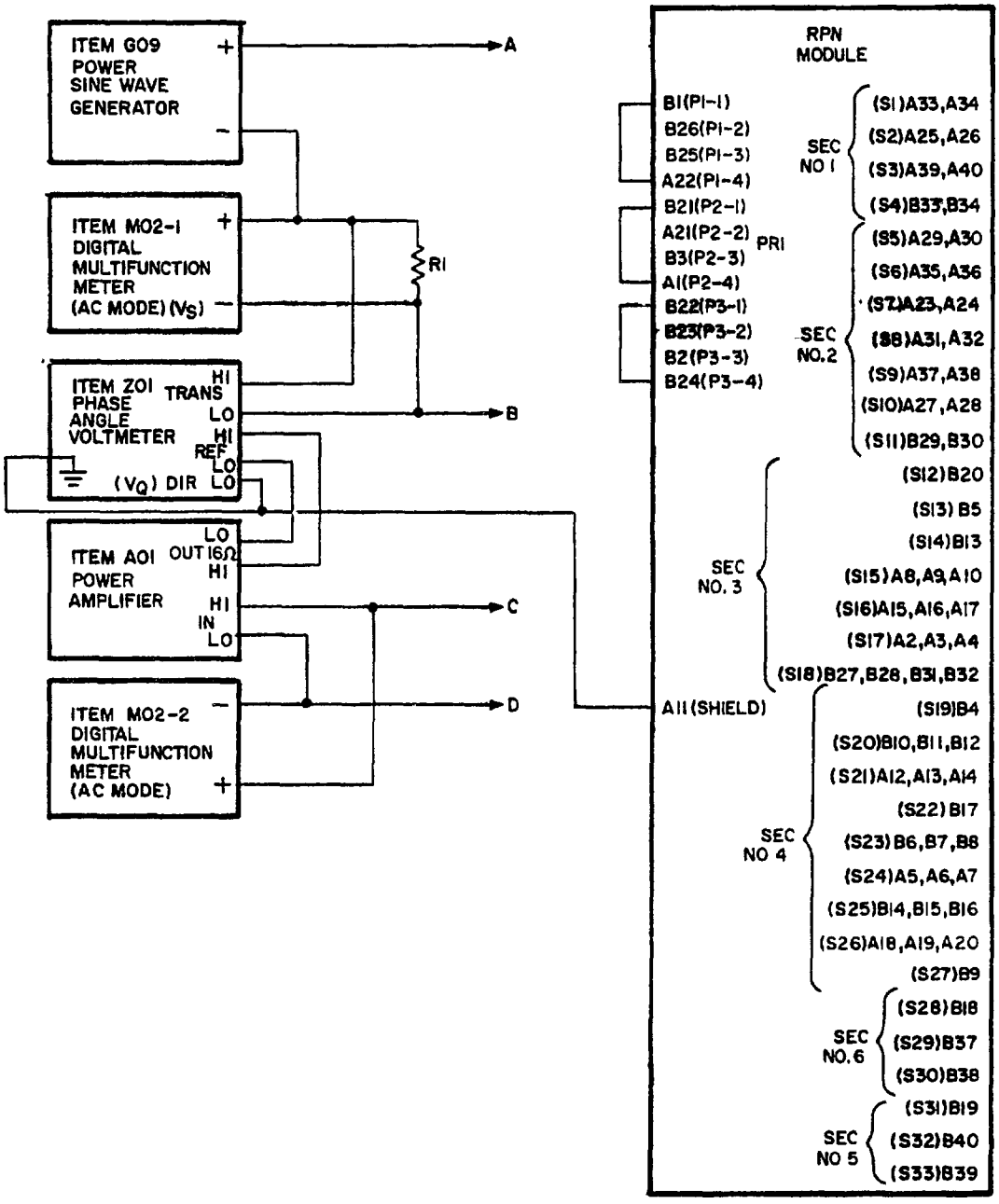
FIGURE 9. Transformation ratio(loaded) test circuit for table V - Continued.

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TABLE VI. Commutating inductance test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	Commutating inductance	A34	B34	A33	B33	0	36	µH	10
F2		A26	B34	A25	B33				
F3		A40	B34	A39	B33				
F4		A24	B30	A23	B29				
F5		A36	B30	A35	B29				
F6		A30	B30	A29	B29		36		
F7		A2	B28	A3 A4	B27 B31		4.7		
F8		A15	B28	A16 A17	B27 B31				
F9		A8	B28	A9 A10	B27 B31				
F10		A12	B10	A13 A14	B11 B12				
F11		A5	B6	A6 A7	B7 B8				
F12	Commutating inductance	A18	B14	A19 A20	B15 B16	0	4.7	µH	10

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* FIGURE 10. Commutating inductance test circuit for table VI.

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NOTES:
1. R1 = 0.1Ω ±1%, 10 W.
2. The circuit lead resistance between the 0.1Ω resistor and test point B connection to the transformer can reduce measurement accuracy. This lead resistance shall not exceed 0.10Ω.
3. Set gain of item A01 to a voltage gain of 20 ±2. Set item G09 to 400 ±2 Hz. Adjust output of item G09 to obtain the specified value of V_S as read on the item M02-1 with test points A and B connected per test condition F1. Zero item Z01 by disconnecting input to item A01 from the module and connecting across R1. Adjust reference level and then with the function switch in the 90° position, adjust for a quadrature voltage null using the degrees knob. Reconnect item A01 input from the module, start tests of table VI. Record V_{IN}, V_Q and V_S. Calculate commutating inductance using the following formula:

$$L(uH) = \frac{V_{IN} \times V_Q}{2.512 \times 10^{-2} V_S^2}$$

The value V_Q is read across 0.1Ω resistor with the phase angle voltmeter function switch in the 90° position.
4. For test conditions F1-F6, V_S shall be adjusted to 0.207 Vrms ±3%.
5. For test conditions F7-F12, V_S shall be adjusted to 0.425 Vrms ±3%.

* FIGURE 10. Commutating inductance test circuit for table VI - Continued.

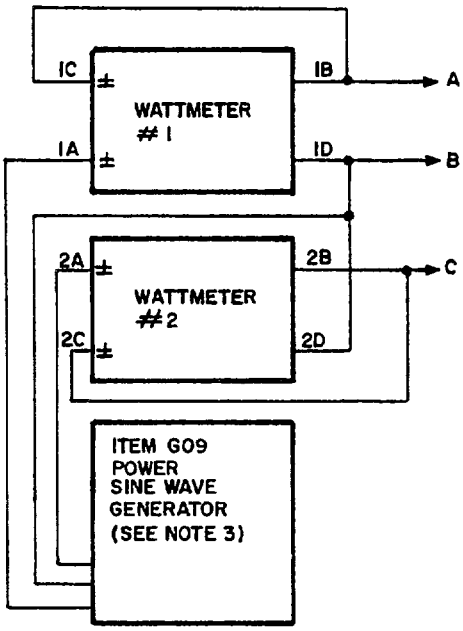
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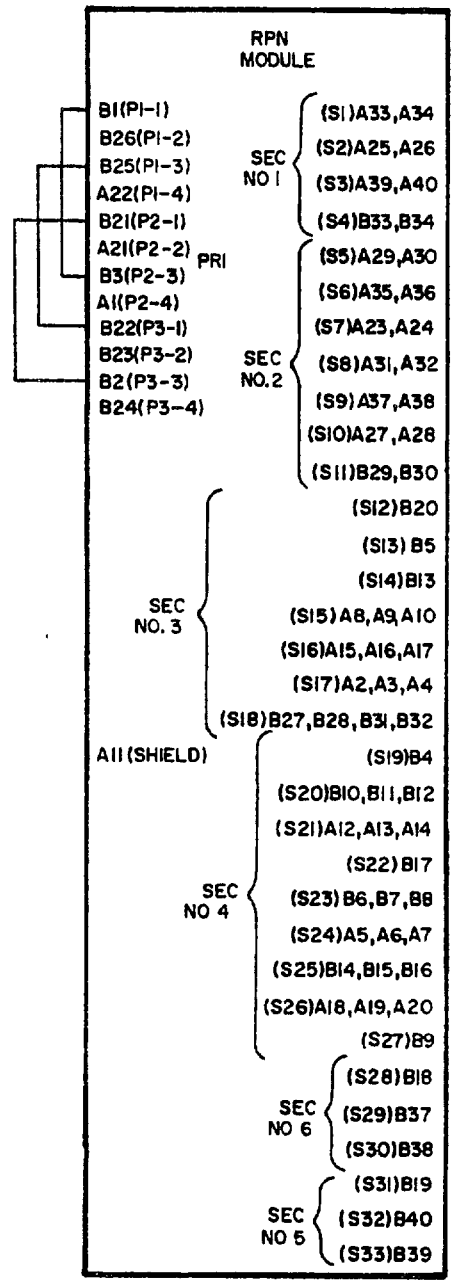
TABLE VII. Core loss test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1a	Core loss	B1	B2	B22	0	See Note	W	11
G1b	Core loss	B2	B22	B1	0	See Note	W	11
G1c	Core loss	B22	B1	B2	0	See Note	W	11

* NOTE: In each measurement, record the power readings algebraically (that is, if the reading of one wattmeter is negative, its potential connections must be reversed and its reading subtracted from the readings of the other wattmeter). Calculate the average of the three readings obtained. The calculated value shall be a maximum of 16 W.

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NOTES
1. Current element terminals 1A, 1B, 2A, and 2B.
2. Voltage element terminals 1C, 1D, 2C, and 2D.
3 Item G09 setup.
115 Vac ±2% (line-to-line).
400 Hz ±2%, 3 phase.

★ FIGURE 11. Core loss test circuit for table VII.

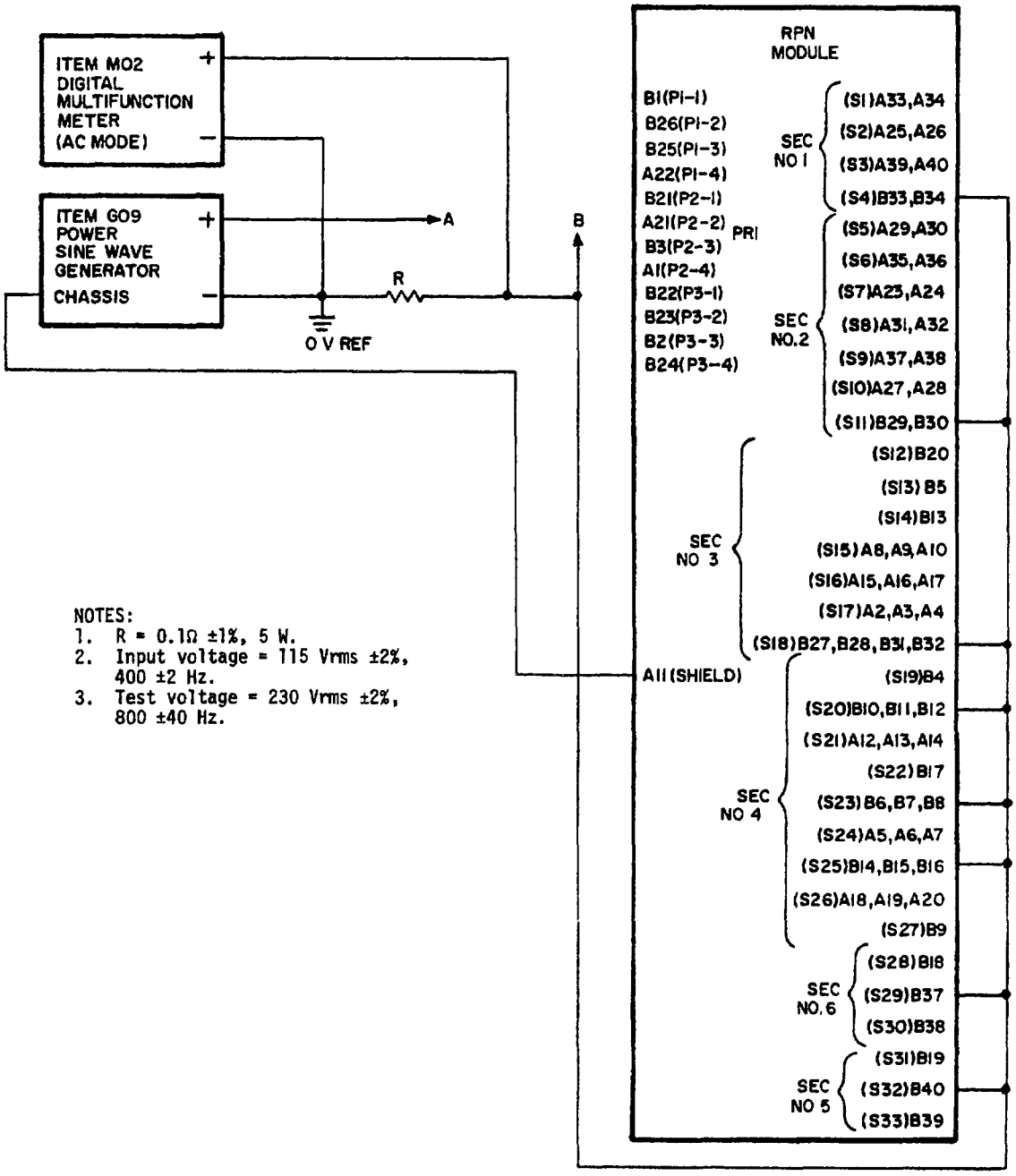
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TABLE VIII. Induced voltage (current) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Induced voltage	B1	B26	0	25	%Δ	12
H2	Induced voltage	B22	B23	0	25	%Δ	12
H3	Induced voltage	B21	A21	0	25	%Δ	12

NOTE: Using the input voltage specified in Figure 12, measure the input current using the 0.1 Ω resistor and record the reading. Then apply the test voltage specified in Figure 12 for 10 seconds or less and again record the reading. The second input current measurement shall be within 25 % Δ (delta) of the first.

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NOTES:
1. R = 0.1Ω ±1%, 5 W.
2. Input voltage = 115 Vrms ±2%, 400 ±2 Hz.
3. Test voltage = 230 Vrms ±2%, 800 ±40 Hz.

* FIGURE 12. Induced voltage test circuit for table VIII.

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* TABLE IX. Jumper resistance test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1A1	Jumper resistance, secondary	2-S7	A23	A24	0	0.020	Ω	6
J1A2		1-S2	A25	A26				
J1A3		2-S10	A27	A28				
J1A4		2-S11	B29	B30				
J1A5		2-S8	A31	A32				
J1A6		1-S4	B33	B34				
J1A7		2-S6	A35	A36				
J1A8		2-S9	A37	A38				
J1A9		1-S3	A39	A40				
J1A10		3-S17	A2	A3				
J1A11		3-S17	A2	A4				
J1A12		4-S24	A5	A6				
J1A13		4-S24	A5	A7				
J1A14		3-S15	A8	A9				
J1A15		3-S15	A8	A10				
J1A16		4-S21	A12	A13				
J1A17		4-S21	A12	A14				
J1A18		3-S16	A15	A16				
J1A19	Jumper resistance, secondary	3-S16	A15	A17	0	0.020	Ω	6

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* TABLE IX. Jumper resistance test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1A20	Jumper resistance, secondary	4-S26	A18	A19	0	0.020	Ω	6
J1A21		4-S26	A18	A20				
J1A22		1-S4	B33	B34				
J1A23		4-S23	B6	B7				
J1A24		4-S23	B6	B8				
J1A25		4-S20	B10	B11				
J1A26		4-S20	B10	B12				
J1A27		3-S18	B27	B28				
J1A28		3-S18	B27	B31				
J1A29		3-S18	B27	B32				
J1A30		2-S11	B29	B30				
J1A31		4-S25	B14	B15				
J1A32	Jumper resistance, secondary	4-S25	B14	B16	0	0.020	Ω	6

NOTE: Readings shall be compensated for test circuit resistance tare in excess of 0.010Ω.

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